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Systematic Review and Meta-Analysis

Does Preoperative Anterior Genicular Nerve Cryoneurolysis Improve Early Outcomes of Primary Total Knee Arthroplasty? A Systematic Review and Meta-Analysis

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ABSTRACT

Background: Preoperative cryoneurolysis (CNL) targeting the anterior genicular nerves has gained attention as a minimally invasive technique to reduce pain in patients undergoing primary total knee arthroplasty (TKA). Our objective was to systematically evaluate CNL in TKA, focusing on pain and opioid consumption, patient-reported outcome measures, length of hospital stay (LOS), and adverse events.
Methods: A systematic literature search was conducted in PubMed, Scopus, Web of Science, and Embase for articles published until 2024, without language restrictions. Comparative studies involving adults undergoing TKA, with or without CNL of the genicular nerves, were included. There were two independent reviewers who screened articles and extracted data. Risk of bias was assessed using the Cochrane RoB-2 tool for randomized trials and the Risk Of Bias In Non-randomized Studies - of Interventions for cohort studies. We evaluated the potential conflict of interest (COI) among studies. The Grading of Recommendations, Assessment, Development and Evaluation approach evaluated the certainty of evidence. Meta-analyses employed random-effects models. There were six comparative studies (n = 676 TKAs) that met the inclusion criteria, including two randomized controlled trials.
Results: The meta-analysis showed that preoperative CNL substantially reduced acute postoperative pain (Hedge's g = −0.33; 95% confidence interval [CI] −0.49 to −0.17) and opioid consumption (Hedge's g = −0.26; 95% CI −0.46 to −0.07). The LOS was also shorter in the CNL group (mean difference = 0.63; 95% CI 0.20 to 1.05). Functional outcomes at 3 months were similar between groups (Hedge's g = 0.15; 95% CI −0.05 to 0.35). Adverse events were minimal; no difference in complications between groups. Four studies reported COI with CNL manufacturers.
Conclusions: Preoperative CNL offers short-term benefits for TKA patients: reducing pain, opioid use, and LOS without increasing adverse events. However, substantial financial COI in studies raises bias concerns, limiting its routine recommendation.
Level of Evidence: III.
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Availability of data and materials: The datasets generated and analyzed during the current study are available from the corresponding author upon reasonable request.

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Cryoneurolysis (CNL) is a technique that exposes peripheral nerves to freezing temperatures, inducing neuronal Wallerian degeneration [1]. This process disrupts sensory nerve function through axonal and myelin degeneration while preserving the nerve architecture and sheath [2,3]. The preservation of the nerve structure facilitates axonal regrowth and remyelination within weeks to months [2]. This could potentially reduce long-term adverse events, making CNL a safe option for pain management [4]. The CNL devices feature a probe designed to induce cold temperatures exclusively at the tip of the gauge [5] (Figure 1).

As a minimally invasive and efficient procedure, CNL has gained traction as a method for managing knee pain and reducing postoperative pain in patients undergoing total knee arthroplasty (TKA) [4,5,8–12]. For TKA patients, CNL targets the anterior genicular nerves of the knee [4,5,8–11] (Figure 2). These nerves play a key role in causing pain associated with the surgical incision and soft tissue injury during TKA [13]. Despite the growing interest in using preoperative CNL as an adjunct for pain management in TKA patients, its safety and efficacy remain inconclusive. While some studies suggest that CNL reduces postoperative pain, decreases opioid consumption, and improves functional scores [4,5,8,9,11]. Other studies have reported minimal effects on these outcomes [4,8,11].

To address this uncertainty, we conducted a systematic review and meta-analysis to evaluate the efficacy and safety of preoperative percutaneous genicular nerve CNL in patients undergoing TKA. Specifically, we asked what the influence of CNL is on: 1) pain and opioid consumption, 2) patient-reported outcome measures (PROMs), 3) length of hospital stay (LOS), and 4) adverse events. We hypothesized that CNL would not significantly improve these defined clinical outcomes.

Methods

This systematic review adheres to the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement [14]. The study protocol was submitted to the International Prospective Register of Systematic Reviews and registered under the identifier CRD42024583084. This study did not require institutional review board approval, as it uses publicly available data.

Eligibility Criteria and Search Strategy

The Population, Intervention, Comparison, Outcome, and Study Design framework was used to define the eligibility criteria. The sample population consisted of adults undergoing primary TKA. The intervention of interest in this study was CNL of the genicular nerves. These patients were compared with a control group that did not receive CNL. The dependent variables included postoperative outcomes such as pain, opioid consumption, knee functional outcomes, sleep outcomes, and complications. Only comparative studies, including randomized controlled trials (RCTs) and cohort studies, were included.

The search for relevant literature was conducted across four databases: PubMed, Scopus, Web of Science, and Embase. The following keywords were used: cryoneurolysis, cryoanalgesia, cryoneuroablation, knee arthroplasty, and knee replacement. Articles published up to December 25, 2024, were considered, with no restrictions on language or publication date. The complete search strategy for each database is detailed in Appendix A. Retrieved articles were organized using EndNote version 21.2 (Clarivate, London, United Kingdom), where duplicates were identified and removed. The remaining articles were then imported into Rayyan.ai [15] for screening.

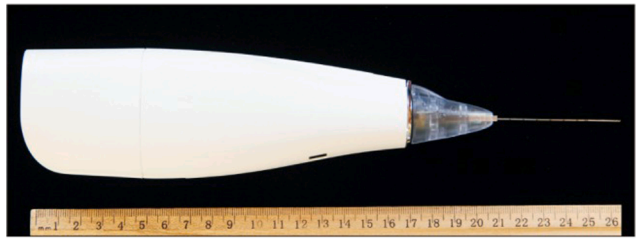

Figure 1. A portable Iovera cryoneurolysis device with a 5.5-cm, 22 gauge cryoprobe. (Iovera Focused Cold Therapy, Myoscience, Freemont, California, USA). This figure is reproduced from *Ilfeld et al.*'s [6] article. Ilfeld BM, et al. [7].

Selection Process

There were two independent reviewers (AN, OB) who screened the titles and abstracts of articles for relevance. Studies that did not meet the inclusion criteria were excluded. Articles deemed potentially relevant were retrieved in full text and assessed for eligibility. This stage also involved two independent reviewers. Disagreements on the inclusion of potentially eligible full text were discussed until a consensus was reached. If a final decision for inclusion could not be agreed on by discussion, a third reviewer (SH) made the final decision.

The initial search in multiple databases returned a total of 266 articles after removing duplicates. After initial screening, 13 studies were eligible for full-text review. Following the full-text review, seven studies were excluded (two case series, five studies with an ineligible study population). Eventually, six studies qualified for the final review, including two randomized controlled trials, one prospective cohort study, and three retrospective cohort studies from 2016 to 2024 [4,5,8–11] (Figure 3).

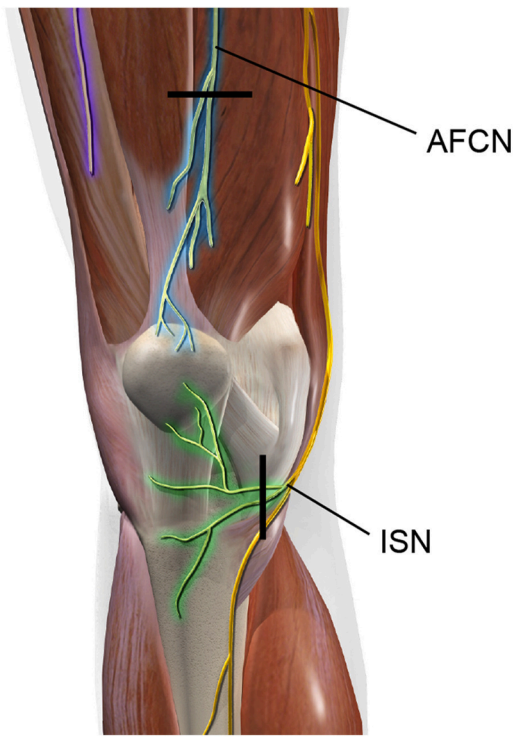

Figure 2. Cryoneurolysis ISN and AFCN treatment lines. AFCN, anterior femoral cutaneous nerve; ISN, infrapatellar branches of the saphenous nerve. This figure is reproduced from *Mihalko et al.*'s [8] article.

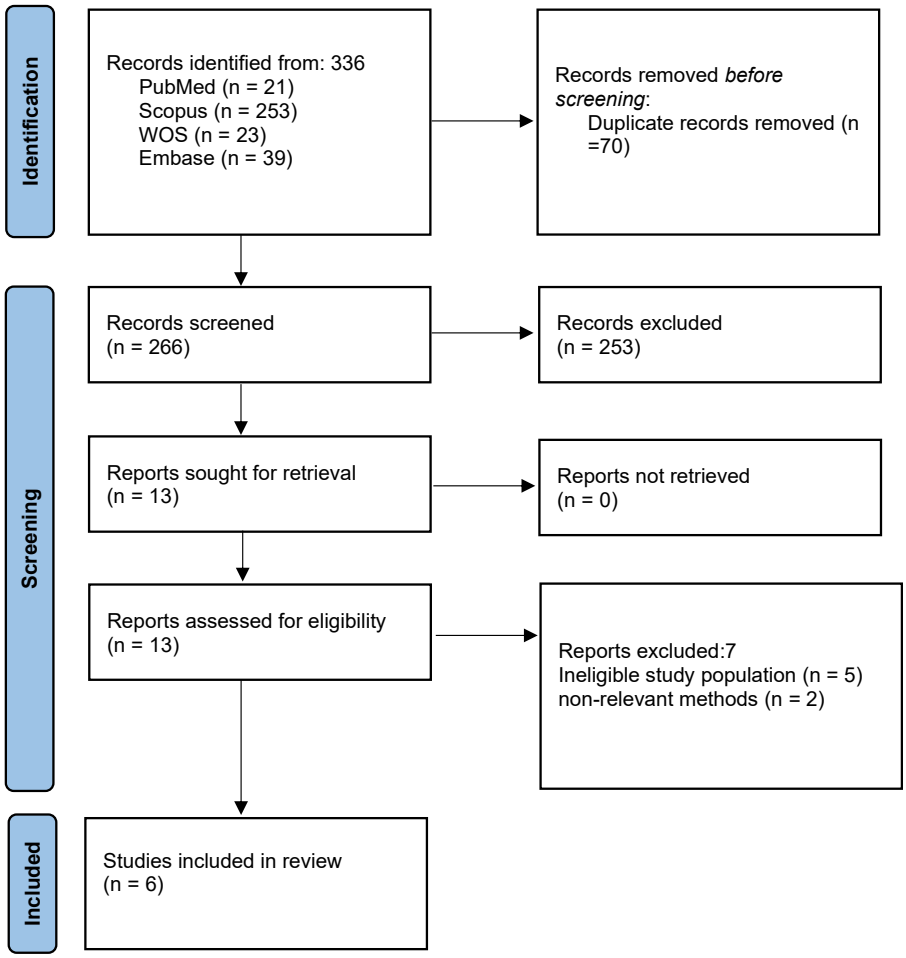

Figure 3. Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) flow diagram outlining the study selection process for this systematic review.

Data Collection Process and Data Items

A detailed extraction sheet in Microsoft Excel version 2,411 (Microsoft, Redmond, Washington) was used for data collection. Data extraction was independently conducted by two reviewers, with any discrepancies resolved through discussion and mutual agreement. The GetData Graph Digitizer software version 2.26 (Software Informer, Chicago, Illinois) was employed to extract data from figures in Mihalko et al. and Mont et al. studies that did not provide numerical data for the Knee injury and Osteoarthritis Outcome Score for Joint Replacement (KOOS JR) [8]. Additionally, data reported as mean ± SE were converted to mean ± SD (SD) if not initially provided in this format. For including the LOS data from the Mont et al. [11] study, which did not provide SDs for the LOS data, we assumed the mean of the SDs from other cohorts in the included studies.

The extracted data included the lead author's name, publication year, study design, geographical setting, baseline characteristics (e.g., sample size, sex, CNL and control cases, age, and body mass index), follow-up period, CNL device and method, postoperative and intraoperative care, and outcome measures such as pain intensity, opioid consumption, Knee injury and Osteoarthritis Outcome Score (KOOS), LOS, and total complications or adverse events.

The study follow-up ranged from 3 weeks to 12 months (Table 1). The total number of patients included in the studies was 676 (254 men, 422 women). The mean age and body mass index of the patients in the studies ranged from 66 and 30.1 years, respectively. The CNL was administered using the Iovera device (Pacira BioSciences, San Diego, California) in all studies except one that used the PainBlocker device (Epimed, Dallas, Texas) [10]. The duration between the intervention and the TKA varied, ranging from the day of operation up to 30 days preoperatively. Superficial genicular nerves, including the infrapatellar branch of the saphenous nerve and the anterior femoral cutaneous nerve, were targeted in all the studies, while one study also targeted deep genicular nerves [11]. For postoperative pain management, acetaminophen and oxycodone were included in the pain regimen for all the studies.

Study Risk of Bias, Certainty of Evidence, and Conflict of Interest Assessments

We evaluated the quality of RCTs and cohort studies using the revised Cochrane risk-of-bias tool for randomized trials (RoB-2) [16] and the Cochrane Risk of Bias in Non-Randomized Studies of Interventions (ROBINS-I) [17], respectively. The results of the risk-of-bias assessments were visualized using the Risk of Bias Visualization tool [18]. In addition, we evaluated the studies for disclosures of financial conflicts of interest (COIs), including funding of trials by companies currently offering CNL technology (Pacira CryoTech, Inc., and Myoscience, Fremont, California) or potential author affiliations and declared COIs.

Table 1
Patients Demographics and Study Characteristics of Included Studies.

Author, Year	Study Design/LoE	Sample Size (CNL, Control)	Men/Women	Age, Mean	BMI, Mean	CNL Device	Target Nerves	Anesthesia	Duration Between CNL and TKA (day)	Postoperative Pain Management Drug	CoI
Mont 2024 [11]	Cohort/III	140 (80, 60)	51/89	66	28.9	Ivora	ISN, AFCN, or deep genicular nerves	NR	0 to 30	NR	Financial CoI
Swisher 2022 [10]	RCT/II	16 (8, 8)	6/10	68	27	Pain blocker	ISN	Single/continuous ACNB, GA	On operation day	Acetaminophen, celecoxib, gabapentin, oxycodone, hydromorphone	Financial CoI
Lung 2022 [4]	Cohort/III	57 (29, 28)	20/37	68	32	Iovera	AFCN, ISN	Single/continuous ACNB	7	NR	Declared no CoI
Urban 2021 [9]	Cohort/III	267 (169, 98)	102/165	67	29.7	Iovera	AFCN, ISN	Single ACNB, GA	9 (8 to 11)	Acetaminophen, meloxicam, tramadol, oxycodone, morphine	Financial CoI
Mihalko 2021 [8]	RCT/II	96 (48, 48)	45/51	65	30.9	Iovera	AFCN, ISN	SA, ACNB, GA	3 to 7	Acetaminophen, gabapentin, meloxicam, tramadol, oxycodone	Financial CoI
Dasa 2016 [5]	Cohort/III	100 (50, 50)	30/70	67	31.6	Iovera	AFCN, ISN	SA + FNB or ACNB	5	Acetaminophen, celecoxib, pregabalin, and oxycodone	Declared no CoI
Total or mean	4 Cohorts and 2 RCTs	676 (384, 292)	(254/422)	66	30.1	5 Iovera, 1 pain blocker	-	-	Range (0 to 30)	-	

LOE, level of evidence; BMI, body mass index; ACNB, adductor canal nerve block; GA, general anesthesia; SA, spinal anesthesia; CNL, cryoneurolysis; ISN, infrapatellar saphenous nerve; AFCN, anterior femoral cutaneous nerve; CoI, conflict of interest; NR, not reported; RCT, randomized controlled trial.

Table 2
Summary of Included Studies Outcomes.

Study	Time Points	Pain	Opioid Consumption	PROMs		Complications		LoS
				Measure	Outcome	CNL	Control	
Mihalko [8]	3 days 2, 6, 12 weeks	Lower pain in CNL group at 3 days and 2 weeks	Lower TME for CNL at 3 days, 6 and 12 weeks	KOOS Jr	Greater KOOS Jr improvement at 2, 6, 12 weeks	2 (pulmonary, DVT)	2 (allergic, neurologic)	NS
Swisher [10]	1, 4 days 1, 2, 3 weeks	N.S for NRS pain score.	N.S for number of opioid tables	-	-	0	0	NS
Mont [11]	1, 2, 3, 4, 5, 6 weeks 2, 3, 4, 5, 6 months	Overall, CNL patients more likely achieved MCID pain reduction for BPI-pain score	Overall, CNL patients S. less likely to report opioid dependence during follow up	KOOS Jr	N.S difference for KOOS Jr MCID achievement	0	0	-
Urban [9]	At discharge 2, 6 weeks	Lower NRS score for CNL during hospitalization	Lower cumulative MME consumption at 2, 6 weeks and total MME consumption at discharge and 6 weeks for CNL	-	-	2 (severe dysesthesia)	0	Lower for CNL
Lung [4]	4, 6 weeks 3, 12 months	-	Lower MME consumption at 4 weeks for CNL group	KOOS Jr	Greater 12 months KOOS Jr improvement for CNL	1 (cardiac)	3 (SSI, pulmonary, cardiac)	NS
Dasa [5]	3, 6, 12 weeks	Greater reduction for PROMIS pain scores for CNL group in 2 and 6 weeks	Lower cumulative MME consumption at 12 weeks	KOOS, WOMAC, OKS	Greater 6, 12 weeks KOOS "Symptoms" improvement for CNL	0	0	Lower for CNL

CNL, cryoneurolysis; N.S, nonsignificant; BPI-SF, Brief Pain Inventory-Short Form; NRS, numeric rating scale; PPI, PROMIS pain intensity; KOOS JR, Knee injury and Osteoarthritis Outcome for Joint Replacement; MME, morphine milligram equivalent; TME, total morphine equivalent; LoS, length of stay; DVT, deep vein thrombosis; SSI, surgical site infection.

The quality of the evidence was further assessed using the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach [19], which examines five key domains: risk of bias (Cochrane RoB-2 or ROBINS-I), inconsistency (I^2 statistic> 50), indirectness, imprecision, and publication bias. A Summary of Findings GRADE profile table was created to present details such as outcomes, participant numbers, effect estimates, and GRADE ratings. Findings were synthesized quantitatively for GRADE assessments. Visualizations were conducted using the GRADEpro GDT tool (available at gradepro.org) [19].

The risk of bias assessment using RoB-2 indicated one RCT with a low risk of bias [10] and another with some concerns. For cohort studies, only one study was assessed as having a low risk of bias [11], while the other three had some concerns for the patient selection domain when evaluated using the ROBINS-I quality assessment tool (Appendix A, B and C). Additionally, four studies disclosed financial CoI [8–11], one study explicitly stated no CoIs [4], and one study did not provide a declaration regarding CoIs [5].

Data Analyses

The meta-analysis was conducted using Comprehensive Meta-Analysis (CMA version 3.3) software. Heterogeneity across studies was assessed using the I^2 statistic, with substantial heterogeneity defined as $I^2 > 50\%$. A random-effects model was employed to provide a more conservative estimate, accounting for variability both within and between studies. Effect sizes for opioid consumption, pain, using Hedges' g, which is well-suited for comparing studies with differing sample sizes and variances for outcome measures. For LOS, the mean difference was calculated as an effect size. A Hedge's g value greater than 0.2 indicates a small effect, greater than 0.5 indicates a medium effect, and greater than 0.8 indicates a large effect. The 0.05 threshold for *P*-value was used to determine statistical significance. We did not assess publication bias and sensitivity analysis due to the limited number of articles in our analysis.

Results

Pain and Opioid Consumption

All six studies compared opioid consumption between groups [4,5,8–11] (Table 2). Most studies indicated a reduction in opioid consumption during their follow-up periods. There were three studies included in the meta-analysis on total opioid consumption at 6 weeks, which was the most commonly reported time point [4,8,9]. The meta-analysis revealed that opioid consumption was lower in the CNL group than the control group with a relatively small effect (Hedge's $g = -0.26$; 95% confidence interval [CI] [−0.46 to −0.07]; $P = 0.01$; $I^2 = 0$) (Figure 4). High-certainty evidence demonstrates that CNL provides a meaningful reduction in acute postoperative pain, with no substantial concerns across GRADE domains (Table 3).

There were five studies that compared pain scores between both groups, including two RCTs and three cohort studies [5,8–11] (Table 2). Overall, studies reported reduced pain most dominantly until 6 weeks of follow-up [5,8,11]. There were five studies with data on early pain scores (up to 2 weeks postoperatively) that were included in the meta-analysis [5,8–11]. Patients in the CNL group experienced lower pain scores than those in the control group (Hedge's $g = -0.33$; 95% CI [−0.49 to −0.17]; $P < 0.001$; $I^2 = 0$) (Figure 5). The GRADE certainty assessment indicated moderate certainty for this outcome domain (Table 3).

Patient-Reported Outcome Measures

There were four studies that reported PROMs, including one RCT and three cohort studies [4,5,8,11] (Table 2). Overall, studies did not report superior outcomes for the KOOS, Western Ontario and McMaster Universities Osteoarthritis Index, Oxford Knee Score, and KOOS Jr overall and subgroup scores. However, there was a trend toward more favorable scores in CNL group with greater KOOS improvement from baseline [4,5,8,11]. Meta-analysis was conducted to evaluate KOOS improvement from baseline at the 3-month endpoint, which was the most commonly reported time point. Four studies were included: three studies reported KOOS Jr [4,8,11], and one study reported KOOS [5]. The meta-analysis indicated that KOOS improvement was comparable between groups at the 3-month endpoint (Hedges' $g = 0.15$; 95% CI [-0.05 to 0.35]; $P = 0.14$; $I^2 = 0\%$) (Figure 6). Low-certainty evidence indicates that CNL may help improve PROMs (Table 3).

Length of Stay

The LOS was reported in five studies, one RCT, and four cohorts (Table 2). All of the studies reported a lower LOS in the CNL group,

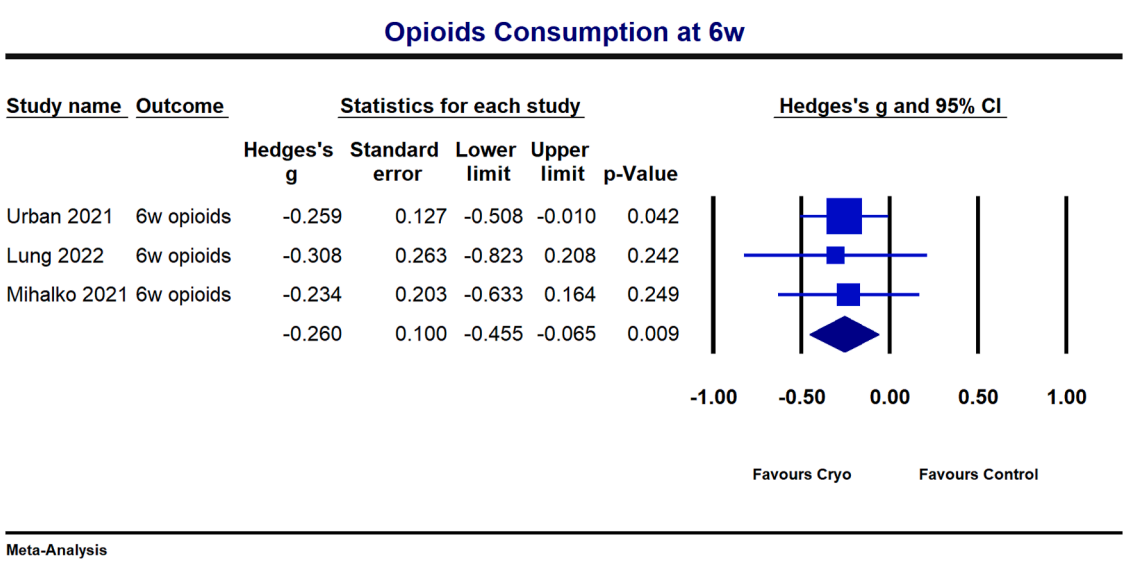

Figure 4. Forest plot of the meta-analysis comparing total opioid consumption between the cryoneurolysis and control groups at 6 weeks postoperatively.

Table 3
Outcomes of Preoperative Cryoneurolysis Compared to Control Group in Patients Who Undergoing Total Knee Arthroplasty.

Certainty Assessment							Summary of Findings				
Participants Follow-Up	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall Certainty of Evidence	Study Event Rates (%)		Relative effect (95% CI)	Anticipated Absolute Effects	
							With Control	With CNL		Risk with Control	Risk Difference with CNL
Acute postoperative pain (follow-Up: 2 weeks; assessed with: numerical pain scores; scale from: 0 to 10)											
619 (5 Nonrandomized studies)[a]	Not serious[b]	Not serious	Not serious	Not serious	None	⊕⊕⊕⊕ High[b]	264	355	-	264	Hedges' *g* **0.33 g lower** (0.49 lower to 0.17 lower)[c]
Mid-term opioid consumption (follow-up: mean 6 weeks; assessed with: milligram morphine equivalents/total morphine equivalents)											
420 (3 Nonrandomized studies)[d]	Not serious[b]	Not serious	Not Serious	Serious[e]	None	⊕⊕⊕○ Moderate[b,e]	174	246	-	174	Hedges' *g* **0.26 g lower** (0.46 lower to 0.07 lower)[c]
Knee Function Score (follow-up: mean 3 months; assessed with: total KOOS and KOOS Jr scores; scale from: 0 to 100)											
393 (4 Non-randomized studies)[d]	Not serious[b]	Not Serious	Not Serious	Very serious[f]	None	⊕⊕○○ Low[b,f]	186	207	-	186	Hedges' *g* **0.15 g higher** (0.05 lower to 0.35 higher)[c]
Length of stay (assessed with: days of hospitalization)											
580 (5 Nonrandomized studies)[d]	Not serious[b]	Not serious[g]	Not Serious	Not Serious	None	⊕⊕⊕⊕ High[b,g]	244	336	-	244	MD **0.63 Days lower** (1.05 Lower to 0.2 Lower)

CI, confidence interval; MD, mean difference.
[a] Two RCTs was included.
[b] There were some concerns about selection bias, but these were not serious enough to downgrade this domain.
[c] We assumed a lower limit of at least 0.1g for a minimal clinically meaningful effect.
[d] One RCT was included.
[e] The calculated confidence interval for the effect size was wide, and one of the limits indicated no effect.
[f] The calculated confidence interval for the effect size was wide and encompassed negative effects, no effect, and positive effects, making the results imprecise.
[g] There was concerning heterogeneity, with $I^2 = 71\%$, however all of studies reported a reduced LoS for CNL, therefore we assumed not serious inconsistency.

and in two studies, this difference was significant. All five studies were included in the meta-analysis [4,5,9–11]. The meta-analysis found that the CNL group had a shorter LOS than the control group (mean difference = −0.63; 95% CI = [−1.05 to −0.20]; P = 0.004; I^2 = 71.6%) (Figure 7). High-certainty evidence indicates that CNL may help improve LOS (Table 3).

Complications and Adverse Events

All studies reported the incidence of complications or adverse events [4,5,8–11] (Table 2). There were three studies that reported no complications or adverse events [5,10,11]. The CNL group had a comparable overall serious adverse event rate to the control group (1.3 for CNL versus 1.7% for the control). Regarding CNL-induced complications, Urban et al. [9] reported only two cases of severe dysesthesia (1.2%).

Discussion

Cryoneurolysis has emerged as a promising, minimally invasive technique for managing postoperative pain in TKA by targeting nerve degeneration while preserving nerve architecture; yet conflicting evidence regarding its efficacy and safety highlights gaps in current knowledge that this systematic review and meta-analysis aimed to address. Our findings, supported by a moderate to high level of certainty, suggest that preoperative genicular nerve CNL improves short-term postoperative outcomes for primary TKA, including reductions in acute pain, opioid use, and LOS, without increasing adverse events, and contributes to improved functional outcomes.

Pain and Opioid Consumption

Cryoneurolysis reduced short-term pain and opioid consumption, while its impact on pain in mid- to long-term time points was inconsistent [5,8]. This discrepancy may reflect TKA's natural recovery process, where pain typically decreases over time, reducing the relative effect of CNL [20–23]. Additionally, the lower overall opioid use in the CNL group, wider early range of motion, and more early distance walked may paradoxically lead to increased activity-related discomfort and a subsequent slight increase in pain and opioid consumption a few weeks postoperatively [4,9,11]. In addition, the timing between CNL and TKA may play a role in early postoperative benefits, warranting further studies to determine the most effective interval for intervention.

Notably, the Non-Opioids Prevent Addiction in the Nation Act recently mandated separate Medicare reimbursement for qualified nonopioid pain relief treatments, including iovera, in hospital outpatient, and ambulatory surgery settings [24]. This policy aims to expand access to alternative treatments and reduce opioid use, potentially leading to the routine use of CNL in practice.

Patient-Reported Outcome Measures

Regarding PROMs, the meta-analysis results indicated no substantial impact of CNL. However, studies showed a trend toward better functional outcomes for the CNL group in mid-term follow-up [4,11]. The early analgesic properties, combined with the impact of CNL on early extended knee range of motion [9] and easier postoperative ambulation [4,11], likely contributed to the slightly superior outcomes observed during the follow-up [25]. Furthermore, as mentioned, the improvement in subjective pain scores and the nonsignificant impact on PROMs, combined with the unblinded nature of most included studies, raises concerns about a potential placebo effect of CNL. Future double-blinded

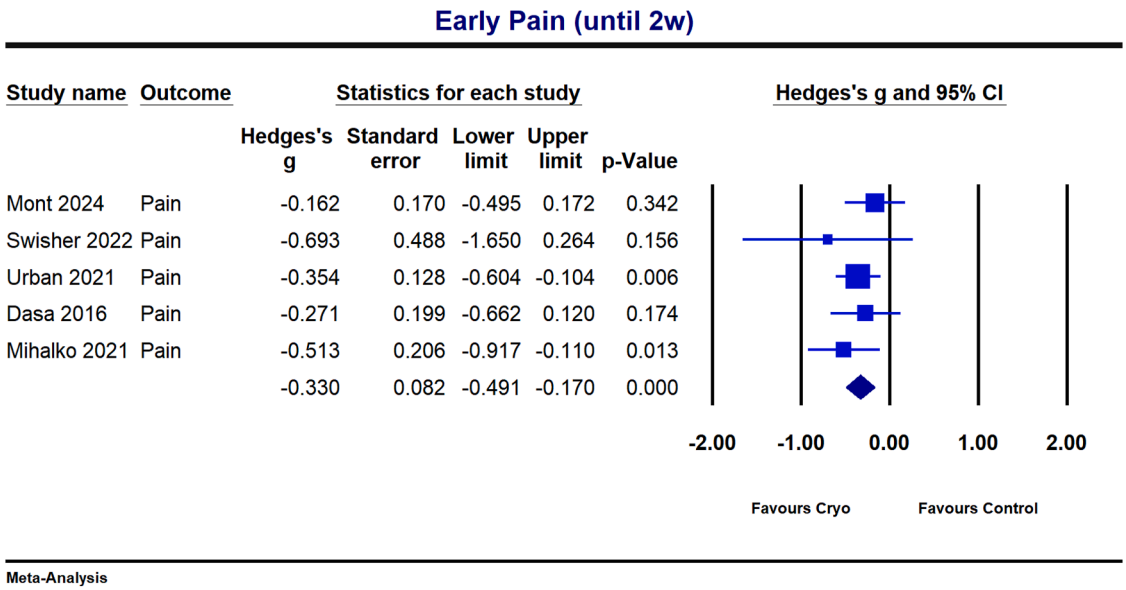

Figure 5. Forest plot of the meta-analysis comparing postoperative pain scores between the cryoneurolysis and control groups during the first 2 weeks.

RCTs are warranted to determine whether CNL truly has an impact on clinical outcomes.

Length of Stay

Cryoneurolysis reduced the LOS, with all included studies reporting shorter LOS for the CNL group. This improvement is likely due to a combination of factors, including lower opioid consumption, better pain management, and subsequently improved ambulation in the CNL group [4,5,9–11]. Effective pain control not only facilitates earlier mobilization but also may reduce complications associated with prolonged hospital stays, such as deep vein thrombosis or infections [8]. Moreover, shorter LOS can result in lower overall health care costs, reduced use of hospital resources and consumables, and increased patient satisfaction due to a more efficient recovery process [11].

Complications and Adverse Events

Adverse events associated with CNL were minimal, with minor symptoms such as numbness or tingling resolving quickly and not interfering with daily activities [9]. Broader applications, including peripheral and lumbar medial branch nerve treatments, reported a low complication rate of approximately 1.1%, including skin infections, bruising, swelling, nerve pain, dysesthesia, or muscle cramps [26,27]. Nearly 97% of patients in clinical studies experienced no persistent pain or dysesthesias after the procedure [26]. Additionally, the reduced opioid consumption observed with CNL may contribute to fewer opioid-related complications, such as respiratory depression, falls, and increased health care costs [28,29]. Compared to other pharmacologic interventions, CNL stands out as a relatively safe and minimally invasive option that could enhance postoperative pain management in TKA. Despite these clinical advantages, the effectiveness of CNL remains

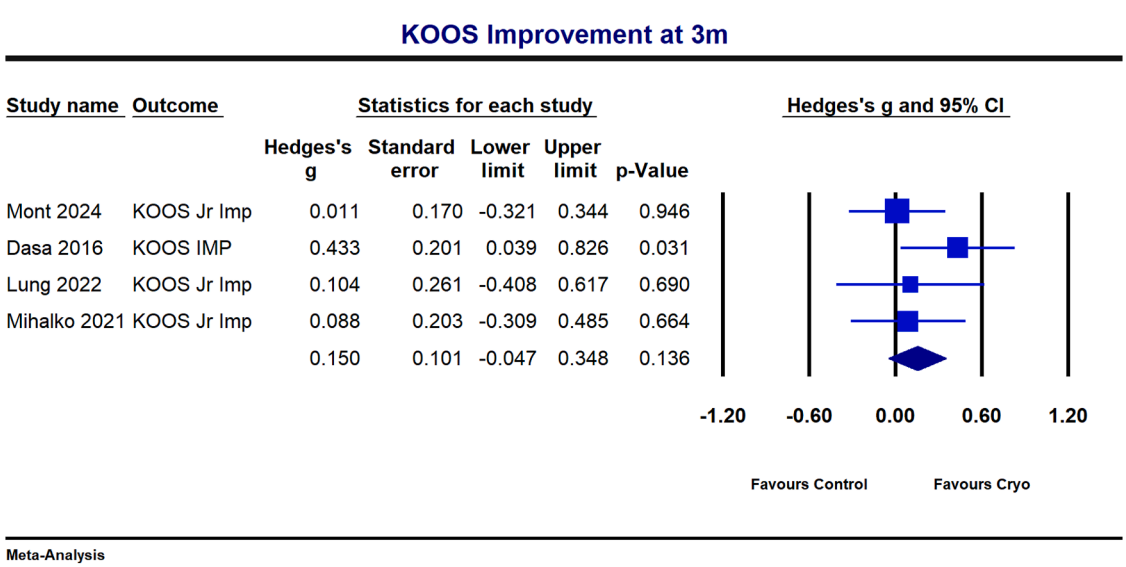

Figure 6. Forest plot of the meta-analysis comparing Knee Injury and Osteoarthritis Outcome Score (KOOS) improvement from baseline at 3 months between the cryoneurolysis and control groups.

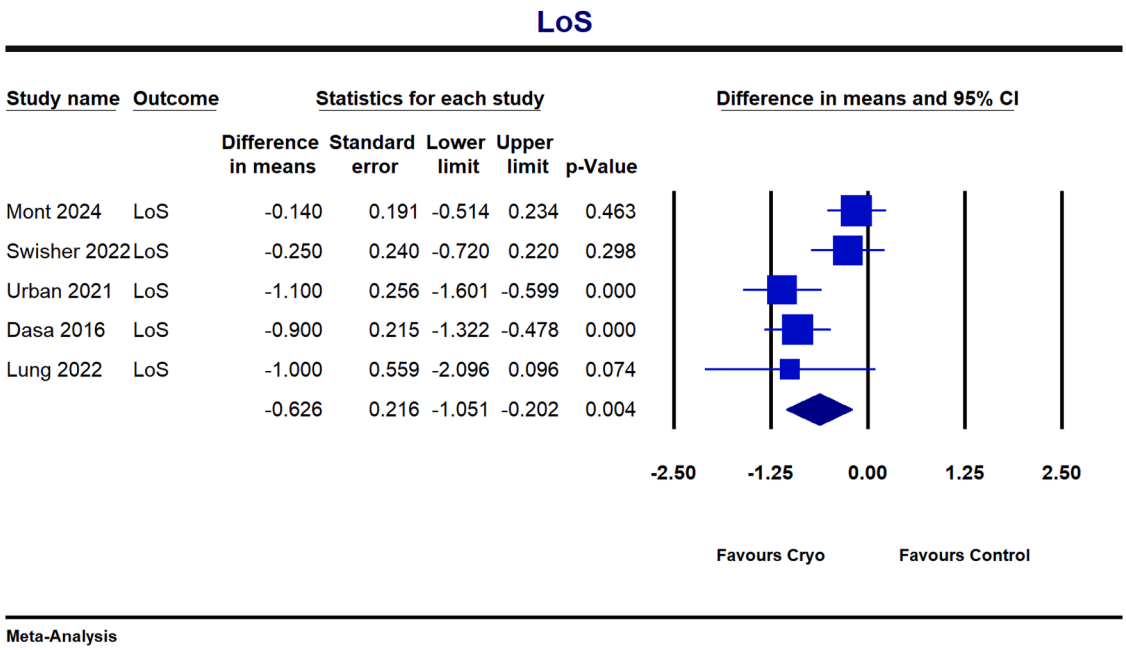

Figure 7. Forest plot of the meta-analysis comparing the length of hospital stay (LOS) between the cryoneurolysis and control groups.

questionable due to the presence of major financial COI. Among the included studies, only one explicitly declared the absence of COI. Previous research has demonstrated that funding sources and financial COI in clinical studies are often associated with more favorable outcomes [6]. As a result, the presence of COI bias casts doubt on the reliability of CNLs' reported effectiveness.

Potential Limitations

This study has several potential limitations. Most of the included studies disclosed financial COI, which could introduce major bias and potentially impact the accuracy of the reported results and their interpretation. Additionally, our meta-analysis had a relatively small pooled sample size, which may have resulted in underpowered analyses. Variability in the timing of CNL administration in relation to surgery across studies further complicates the interpretation of results. The use of different assessment tools across studies may have contributed to inconsistencies in outcome measurements. Furthermore, some meta-analyses were limited to a subset of studies due to incomplete or noncomparable data.

Conclusions

Preoperative CNL appears to provide short-term benefits for patients undergoing TKA by reducing acute postoperative pain, opioid consumption, and LOS. This suggests that CNL is a promising adjunct to enhance postoperative recovery without an increased risk of adverse events. However, the presence of financial COI in most studies raises concerns about potential bias, making it premature to recommend the routine use of CNL in clinical practice. Future high-quality research with larger sample sizes and without COI is necessary to determine the true effectiveness of CNL.

CRediT authorship contribution statement

Sina Hajiaghajani: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Poursalehian:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ali N. Samakoosh:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Omid Bahrami:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Christian J. Hecht:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Atul F. Kamath:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

References

[1] Ilfeld BM, Finneran IV JJ. Cryoneurolysis and percutaneous peripheral nerve stimulation to treat acute pain: a narrative review. Anesthesiology 2020;133:1127–49. https://doi.org/10.1097/aln.0000000000003532.
[2] Barnard D. The effects of extreme cold on sensory nerves. Ann R Coll Surg Engl 1980;62:180–7.
[3] Zhou L, Kambin P, Casey KF, Bonner FJ, O'Brien E, Shao Z, et al. Mechanism research of cryoanalgesia. Neurol Res 1995;17:307–11. https://doi.org/10.1080/01616412.1995.11740333.
[4] Lung BE, Karasavvidis T, Sharma AK, Amirhekmat A, Stepanyan H, McMaster W, et al. Cryoneurolysis is a safe, effective modality to improve rehabilitation after total knee arthroplasty. Life (Basel) 2022;12:9. https://doi.org/10.3390/life12091344.
[5] Dasa V, Lensing G, Parsons M, Harris J, Volaufova J, Bliss R. Percutaneous freezing of sensory nerves prior to total knee arthroplasty. Knee 2016;23:523–8. https://doi.org/10.1016/j.knee.2016.01.011.
[6] Ahn R, Woodbridge A, Abraham A, Saba S, Korenstein D, Madden E, et al. Financial ties of principal investigators and randomized controlled trial outcomes: cross sectional study. BMJ 2017;356:i6770. https://doi.org/10.1136/bmj.i6770.
[7] Ilfeld BM, Gabriel RA, Trescot AM. Ultrasound-guided percutaneous cryoneurolysis providing postoperative analgesia lasting many weeks following a single administration: a replacement for continuous peripheral nerve blocks? a case report. Korean J Anesthesiol 2017;70:567–70. ISSN: 2005-6419, https://doi.org/10.4097/kjae.2017.70.5.567.

[8] Mihalko WM, Kerkhof AL, Ford MC, Crockarell JR, Harkess JW, Guyton JL. Cryoneurolysis before total knee arthroplasty in patients with severe osteoarthritis for reduction of postoperative pain and opioid use in a single-center randomized controlled trial. J Arthroplasty 2021;36:1590–8. Elsevier B.V.
[9] Urban JA, Dolesh K, Martin E. A multimodal pain management protocol including preoperative cryoneurolysis for total knee arthroplasty to reduce pain, opioid consumption, and length of stay. Arthroplast Today 2021;10:87–92. https://doi.org/10.1016/j.artd.2021.06.008.
[10] Swisher MW, Ball ST, Gonzales FB, Cidambi KR, Trescot AM, Ilfeld BM. A randomized controlled pilot study using ultrasound-guided percutaneous cryoneurolysis of the infrapatellar branch of the saphenous nerve for analgesia following total knee arthroplasty. Pain Ther 2022;11:1299–307. https://doi.org/10.1007/s40122-022-00427-4.
[11] Mont MA, Lin JH, Spitzer AI, Dasa V, Rivadeneyra A, Rogenmoser D, et al. Cryoneurolysis associated with improved pain, function, and sleep in patients following total knee arthroplasty: use of a new real-world registry. J Arthroplasty 2024;40:92–101.e3. https://doi.org/10.1016/j.arth.2024.06.054.
[12] Roth ZA, Sutton K, Wenende J, Pecka S. Preoperative cryoneurolysis for total knee arthroplasty: a case series. J Perianesth Nurs 2023;38:33–8. https://doi.org/10.1016/j.jopan.2022.03.006.
[13] Horner G, Dellon AL. Innervation of the human knee joint and implications for surgery. Clin Orthop Relat Res 1994;301:221–6.
[14] Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The prisma 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. https://doi.org/10.1136/bmj.n71.
[15] Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan—a web and mobile app for systematic reviews. Syst Rev 2016;5:210. https://doi.org/10.1186/s13643-016-0384-4.
[16] Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. Rob 2: a revised tool for assessing risk of bias in randomised trials. BMJ 2019;366:l4898. https://doi.org/10.1136/bmj.l4898.
[17] Sterne JA, Hernán MA, Reeves BC, Savović J, Berkman ND, Viswanathan M, et al. Robins-i: a tool for assessing risk of bias in non-randomised studies of interventions. BMJ 2016;355:i4919. https://doi.org/10.1136/bmj.i4919.
[18] McGuinness LA, Higgins JPT. Risk-of-bias visualization (robvis): an r package and shiny web app for visualizing risk-of-bias assessments. Res Synth Methods 2021;12:55–61. https://doi.org/10.1002/jrsm.1411.
[19] Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. Grade: an emerging consensus on rating quality of evidence and strength of recommendations. BMJ 2008;336:924–6. https://doi.org/10.1136/bmj.39489.470347.AD.
[20] Mekkawy KL, Zhang B, Wenzel A, Harris AB, Khanuja HS, Sterling RS, et al. Mapping the course to recovery: a prospective study on the anatomic distribution of early postoperative pain after total knee arthroplasty. Arthroplasty 2023;5:37. https://doi.org/10.1186/s42836-023-00194-3.
[21] Bittman RW, Peters GL, Newsome JM, Friedberg EB, Mitchell JW, Knight JM, et al. Percutaneous image-guided cryoneurolysis. Am J Roentgenol 2018;210:454–65. https://doi.org/10.2214/ajr.17.18452.
[22] Nilsdotter AK, Toksvig-Larsen S, Roos EM. Knee arthroplasty: are patients' expectations fulfilled? Acta Orthop 2009;80:55–61. https://doi.org/10.1080/17453670902805007.
[23] Brander VA, Stulberg SD, Adams AD, Harden RN, Bruehl S, Stanos SP, et al. Ranawat award paper: predicting total knee replacement pain: a prospective, observational study. Clin Orthop Relat Res 2003;416:27–36.
[24] Boufadel P, Fares MY, Daher M, Mathur A, Saunders P, Abboud JA, et al. Health policy challenges and reforms: critical updates for orthopaedic surgeons. JBJS Rev 2025;13:2. https://doi.org/10.2106/jbjs.Rvw.24.00192.
[25] Ha CW, Park YB, Song YS, Kim JH, Park YG. Increased range of motion is important for functional outcome and satisfaction after total knee arthroplasty in asian patients. J Arthroplasty 2016;31:1199–203. https://doi.org/10.1016/j.arth.2015.12.018.
[26] Winston P, MacRae F, Rajapakshe S, Morrissey I, Boissonnault È, Vincent D, et al. Analysis of adverse effects of cryoneurolysis for the treatment of spasticity. Am J Phys Med Rehabil 2023;102:11.
[27] Biel E, Aroke EN, Maye J, Zhang SJ. The applications of cryoneurolysis for acute and chronic pain management. Pain Pract 2023;23:204–15. https://doi.org/10.1111/papr.13182.
[28] Trasolini NA, McKnight BM, Dorr LD. The opioid crisis and the orthopedic surgeon. J Arthroplasty 2018;33:3379–33782.e1. https://doi.org/10.1016/j.arth.2018.07.002.
[29] Oderda GM, Said Q, Evans RS, Stoddard GJ, Lloyd J, Jackson K, et al. Opioid-related adverse drug events in surgical hospitalizations: impact on costs and length of stay. Ann Pharmacother 2007;41:400–7. https://doi.org/10.1345/aph.1H386.

Appendix A

Appendix A
Systematic Search Protocol (Search Date: December 25, 2024).

Data Source	Search Strategy	Results
PubMed	("cryoneurolysis"[All Fields] OR "cryoanalgesia"[All Fields] OR "cryoneuroablation"[All Fields]) AND ("knee arthroplasty"[All Fields] OR "knee replacement"[All Fields] OR "arthroplasty, replacement, knee"[MeSH Terms])	21
Scopus	ALL (cryoneurolysis OR cryoanalgesia OR cryoneuroablation) AND ALL ("knee arthroplasty" OR "knee replacement")	253
WOS	ALL= (cryoneurolysis OR cryoanalgesia OR cryoneuroablation) AND ALL= ("knee arthroplasty" OR "knee replacement")	23
Embase	('cryoneurolysis'/exp OR cryoneurolysis OR 'cryoanalgesia'/exp OR cryoanalgesia OR cryoneuroablation) AND ('knee arthroplasty'/exp OR 'knee arthroplasty' OR 'knee replacement'/exp OR 'knee replacement')	39
Total	-	336
Duplicate	-	70
Reviewed	-	266
Included	-	6

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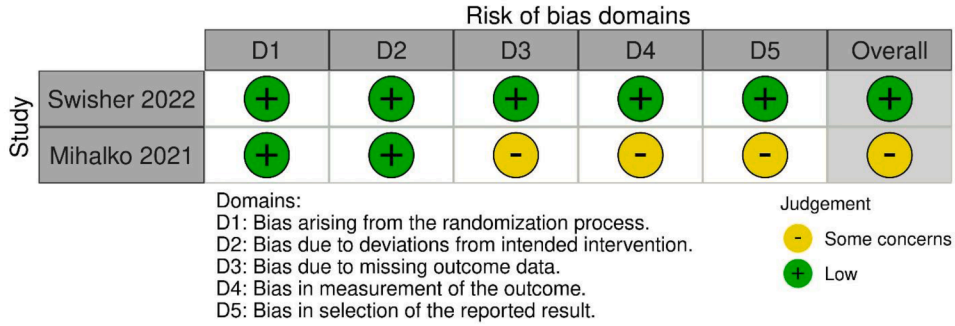
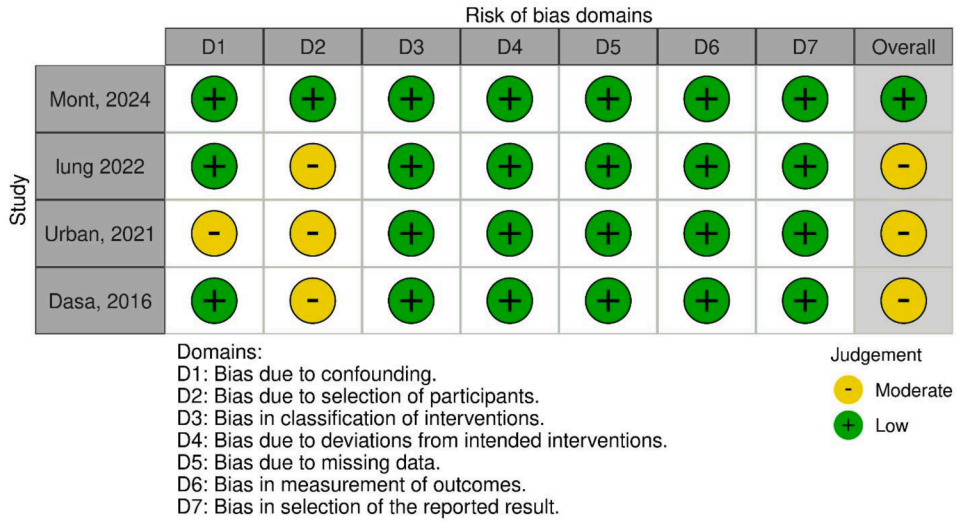
Appendix B
Study Results.

Author	Groups	Pain Score	KOOS (3 months)	Opioids Consumption	LoS (day)	Total Adverse Events
Mont [11]	CNL	BPI-SF/1 week 3.82 ± 5.10	Jr 68.96 ± 32.2	-	0.72 ± 1.09	0
	Control	4.64 ± 4.96	68.12 ± 26	-	0.86 ± 1.15	0
Swisher [10]	CNL	NRS/2 weeks 2.38 ± 1.11	-	-	1.38 ± 0.48	0
	Control	3.86 ± 2.63	-	-	1.63 ± 0.48	0
Lung [4]	CNL	-	Jr 66.4 ± 14.8	MMEs 6 weeks:154 ± 427	2.5 ± 0.8	1 (cardiac)
	Control	-	62 ± 11.9	MMEs: 6 weeks:289 ± 439	3.5 ± 2.9	3 (SSI, pulmonary, cardiac)
Urban [9]	CNL	NRS/until discharge 3.06 ± 2.49	-	MMEs: 6 weeks: 34 ± 10.75	1.42 ± 1.92	2 (Severe dysesthesia induced by CNL)
	Control	3.92 ± 2.3	-	MMEs: 6 weeks: 87 ± 27.75	2.52 ± 0.21	0
Mihalko [8]	CNL	-	Jr 16.0 ± 3.25	TME 6 weeks: 4.2 ± 3.5	-	2 (shortness of breath, DVT)
	Control	-	14.1 ± 2.93	6 weeks: 5.9 ± 4.2	-	2 (allergic reaction, neurologic symptoms)
Dasa [5]	CNL	PPI/2 weeks 5.7 ± 2.3	KOOS total 67.57 ± 23.82	-	0.8 ± 1.14	0
	Control	2 Weeks: 6.4 ± 2.8	59.48 ± 24.39	-	1.7 ± 1.01	0

CNL, cryoneurolysis; BPI-SF, Brief Pain Inventory-Short Form; NRS, numeric rating scale; PPI, PROMIS pain intensity; KOOS JR, Knee injury and Osteoarthritis Outcome for Joint Replacement; MME, morphine milligram equivalent; TME, total morphine equivalent; LoS, length of stay; DVT, deep vein thrombosis; SSI, surgical site infection.





Appendix C. Risk of bias of individual studies per domain.